

TODO: Title of the multimodal cancer prediction study

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Abstract

Background: [TODO: State the cancer prediction task, motivation, study objective, and model name.]
Results: [TODO: Summarize the completed comparisons in Tables 2 and 3, if Table 3 is retained; insert only documented results.]
Conclusions: [TODO: State conclusions supported by the completed comparisons. Do not insert a performance or novelty claim yet.]

Keywords: Multimodal learning; cancer prediction; textual data; genomic data; multimodal fusion

1 Background

[TODO: Introduce the clinical prediction task and the motivation for combining textual and genomic information.]

The Cancer Genome Atlas (TCGA) provides the data-source context for this draft [?]. [TODO: Describe the specific data used and the research question.]

The baseline reporting framework uses Table 1 of Song et al. [?] as the planned source of published comparison values. [TODO: Add the study rationale and a factual description of Ours once the method has been specified.]

2 Methods

2.1 Data and input checklist

- Cohort columns: BLCA, BRCA, GBMLGG, LUAD, UCEC, CRAD, HNSC, and STAD. [TODO: Record cohort definitions, sample counts, inclusion criteria, and data versions.]
- Two-modality inputs: text and genomics. [TODO: Record data fields, preprocessing, and feature representations.]
- Optional three-modality inputs: text, genomics, and [TODO: third modality and its representation].
- Prediction target: [TODO: survival or classification, outcome definition, labels, and prediction time point if applicable].

2.2 Baseline comparison checklist

- Table 2: Cross-Attn Fusion, Early Fusion, Late Fusion, and Ours.
- Table 3, if retained: LIMOE, MAGGate, MulT, TF, Cross-Attn Fusion, Early Fusion, Late Fusion, and Ours.

- Ours: [TODO: model name, existing implementation identifier, and a short factual method description].

Bibliographic slots are reserved for Cross-Attn Fusion, Early Fusion, and Late Fusion [? ? ?], and for LIMOE, MAGGate, and Transformer Fusion (TF) [? ? ?]. MulT has a candidate original reference [?]. [TODO: Resolve each TODO-CITE and confirm the mapping from each table label to its cited method.]

2.3 Evaluation and filling checklist

- Validation for Ours: five-fold evaluation. [TODO: Record split files, fold identifiers, and their relationship to the source study.]
- Per-cohort reporting: fill the eight cohort columns in Table 2 and, if retained, Table 3.
- Metric: [TODO: full metric name, abbreviation, implementation, and whether higher or lower values are better].
- Summary within each cohort: [TODO: aggregation across folds and the uncertainty measure reported beside the point estimate].
- Overall: [TODO: aggregation definition, weighting, and source or calculation record for each row].
- Ours result records: [TODO: paths to existing per-fold results, configurations, and the checkpoint-selection rule].

2.4 Provenance of comparison values

Non-Ours values in Tables 2 and 3 are planned to be transcribed from Table 1 of Song et al. [?]. These are literature-reported results; they are not experiments reproduced by the authors of this study. Ours rows are reserved for the authors’ own experimental results.

[TODO: Record the source row, column, and setting for each transcribed value. Record the calculation used for any derived Overall entry.] All numerical entries remain unfilled in this draft.

2.5 Prototype-guided compensation for missing modalities

Scope. This subsection specifies the inference-time method that is evaluated as the third strategy in Table 1. It is a non-parametric compensation guided by the cached whole-slide image (WSI) prototypes of each patient: a missing modality is filled with features taken from the most similar training patient, shrunk towards the training mean by a setting-specific coefficient, and compensation can be switched off for a setting. Nothing is trained in this study. The survival networks are frozen (one previously trained NPJ-C model per cohort and seed, 25 in total), and the method involves no learnable prototypes, no learned compensation module, no moving-average update, no contrastive or alignment loss and no additional training objective. The experiment covers five TCGA cohorts (BLCA, BRCA, LGG, LUAD and UCEC), three modalities and five seeds; the seeds index independently trained networks and are not cross-validation folds. The equations below restate the executed computation and are not a derivation of optimality.

Modality availability. Let c denote a cohort, i a query patient, j a candidate donor, $m \in \{W, R, T\}$ the WSI, RNA and text modalities, g an evaluation setting and $\mathcal{I}_c$ the training patients of cohort c . The cached feature array of modality m before projection is $\mathbf{X}_i^m$, of size 128×1536 for the WSI, 2048×256 for RNA and 200×768 for text. With $n_i^m \in \{0, 1\}$ indicating that

modality m is naturally available and $b_g^m \in \{0, 1\}$ the artificial mask of setting g , the availability seen by the model is

$$v_{i,g}^m = n_i^m b_g^m, \quad (b_g^W, b_g^R, b_g^T) = \begin{cases} (1, 1, 1), & g = \text{none}, \\ (1, 0, 1), & g = \text{rna_100}, \\ (1, 1, 0), & g = \text{text_100}, \\ (1, 0, 0), & g = \text{both_100}. \end{cases} \quad (1)$$

By Eq. (1) the WSI is never masked, and the setting **none** adds no artificial mask: naturally missing modalities remain missing, so this setting is not a complete-data condition. The order (W, R, T) is notational; the implementation orders the modality tokens as image, text, RNA.

Retrieval key from valid WSI prototypes. The WSI cache $\mathbf{P}_i = \mathbf{X}_i^W \in \mathbb{R}^{128 \times 1536}$ holds at most 128 rows per patient, namely patient-level cluster centres of patch embeddings or, when fewer rows are available, the available rows followed by zero padding. The number 128 is therefore an upper bound, and not every row is a genuine prototype. The cache is used as stored; no clustering is repeated. Valid rows are identified on the uncentred cache and averaged,

$$a_{ik} = \mathbf{1}[\mathbf{P}_{ik} \neq \mathbf{0}], \quad \bar{\mathbf{p}}_i = \frac{\sum_{k=1}^{128} a_{ik} \mathbf{P}_{ik}}{\sum_{k=1}^{128} a_{ik}} \in \mathbb{R}^{1536}, \quad (2)$$

where the only admissible layout is a non-zero prefix followed by a zero suffix. The mean in Eq. (2) excludes padding from retrieval. The key is then centred with the training mean of the same cohort,

$$\boldsymbol{\mu}_c^W = \frac{1}{|\mathcal{I}_c|} \sum_{j \in \mathcal{I}_c} \bar{\mathbf{p}}_j, \quad \mathbf{q}_i = \bar{\mathbf{p}}_i - \boldsymbol{\mu}_c^W. \quad (3)$$

In Eq. (3) every training patient has the same weight regardless of its number of patches, and validation and test patients never contribute to $\boldsymbol{\mu}_c^W$. The retrieval key $\mathbf{q}_i$ is a single 1536-dimensional vector; the row-aligned cosine over the flattened 128×1536 array used by an earlier protocol is not part of this one.

Similarity, candidates and donor selection. The similarity between query i and training patient j is

$$s_{ij}^W = \frac{\mathbf{q}_i^\top \mathbf{q}_j}{\|\mathbf{q}_i\|_2 \|\mathbf{q}_j\|_2}, \quad s_{ij}^R = \frac{\bar{\mathbf{r}}_i^\top \bar{\mathbf{r}}_j}{\|\bar{\mathbf{r}}_i\|_2 \|\bar{\mathbf{r}}_j\|_2}, \quad s_{ij} = s_{ij}^W + J_{i,g} \alpha s_{ij}^R, \quad (4)$$

where $\bar{\mathbf{r}}_i$ is the token mean of $\mathbf{X}_i^R$ without centring and $\alpha = 1$. The indicator $J_{i,g}$ equals one only if $g = \text{text_100}$, features are retrieved in that setting ($\gamma_{c,g} = 1$ and $\lambda_{c,g} = 1$, both defined below) and the patient’s own RNA is naturally available; otherwise $J_{i,g} = 0$ and no RNA is read. In Eq. (4) a centred key norm or an RNA norm not exceeding 10^{-12} raises an error instead of being regularised. With $\mathcal{M}_{i,g} = \{m \in \{R, T\} : v_{i,g}^m = 0\}$ the modalities to be compensated, the admissible donors are

$$\mathcal{D}_{i,g} = \left\{ j \in \mathcal{I}_c : j \neq i, \quad n_j^m = 1 \quad \forall m \in \mathcal{M}_{i,g}, \quad (J_{i,g} = 1 \Rightarrow n_j^R = n_j^T = 1) \right\}. \quad (5)$$

Equation (5) restricts donors to training patients of the same cohort that naturally possess every modality the query lacks, and it excludes the query itself. When both RNA and text are missing, a donor must possess both, so that one donor supplies both modalities. Validation and test patients never enter the patient bank, survival times and censoring indicators are not

used for retrieval, and an empty candidate set raises an error. The donor is the most similar admissible patient,

$$j^* = \arg \max_{j \in \mathcal{D}_{i,g}} s_{ij}, \quad (6)$$

and ties in Eq. (6) are resolved by taking the first patient in sorted identifier order, which makes retrieval deterministic.

Compensated value. Let $\mathcal{I}_c^m = \{j \in \mathcal{I}_c : n_j^m = 1\}$. The training mean of modality m and the compensated value are

$$\boldsymbol{\mu}_c^m = \frac{1}{|\mathcal{I}_c^m|} \sum_{j \in \mathcal{I}_c^m} \mathbf{X}_j^m, \quad \tilde{\mathbf{X}}_i^m = \boldsymbol{\mu}_c^m + \lambda_{c,g}(\mathbf{X}_{j^*}^m - \boldsymbol{\mu}_c^m), \quad \lambda_{c,g} \in \{0, 1\}. \quad (7)$$

The mean in Eq. (7) is element-wise over complete feature arrays of identical shape. The WSI prototypes act only as the retrieval key: the value returned is the donor’s complete pre-projection feature array of the target modality, and cluster centres are never substituted for RNA or text. The coefficient $\lambda_{c,g}$ is specific to cohort c and setting g . With $\lambda_{c,g} = 1$ the compensated value equals the donor’s original features, selected by Eqs. (4)–(6); with $\lambda_{c,g} = 0$ the donor term vanishes, the value reduces to the training mean and no retrieval is performed. In addition, a binary gate $\gamma_{c,g} \in \{0, 1\}$ determines whether compensation is applied at all in cohort c and setting g : with $\gamma_{c,g} = 0$ the missing modalities remain masked. Both reference strategies are special cases of this formulation, namely no compensation ($\gamma_{c,g} = 0$ throughout) and mean imputation ($\gamma_{c,g} = 1$ and $\lambda_{c,g} = 0$ throughout). The constants α and w (defined below) and the default $\lambda_{c,g} = 1$ were selected on the validation split. **[TODO: Author to add: values of $\lambda_{c,g}$ and $\gamma_{c,g}$ for each cohort and setting (needed to reproduce the identical entries of Table 1).]**

Filling and fusion mask. The model input and the fusion validity flag are

$$\mathbf{X}_{i,\text{in}}^m = \begin{cases} \mathbf{X}_i^m, & v_{i,g}^m = 1 \text{ or } \gamma_{c,g} = 0, \\ \tilde{\mathbf{X}}_i^m, & v_{i,g}^m = 0 \text{ and } \gamma_{c,g} = 1, \end{cases} \quad (8)$$

$$f_i^m = \max \left\{ v_{i,g}^m, \gamma_{c,g} \mathbf{1}[m \in \mathcal{M}_{i,g}] \right\}. \quad (9)$$

By Eq. (10) features that the patient already has are never replaced, and when both modalities are filled with $\lambda_{c,g} = 1$ they come from the same donor j^* . With $\gamma_{c,g} = 0$ the missing slot keeps a placeholder that Eq. (11) marks as invalid, so it is excluded from fusion; with $\gamma_{c,g} = 1$ the filled slot is marked valid. Means, similarities and the compensated value are computed in double precision and cast back to the input precision of the network.

Frozen encoder and fusion. Each modality enters the frozen network through its original interface,

$$\mathbf{z}_i^m = \phi_m(\text{mean}_{\text{token}}(\mathbf{X}_{i,\text{in}}^m)), \quad \mathbf{e}_i^m = \mathbf{z}_i^m + \mathbf{d}_m, \quad \phi_m = \text{Dropout} \circ \text{ReLU} \circ \text{Linear}_m, \quad (10)$$

where the projector ϕ_m and the modality embedding $\mathbf{d}_m$ are frozen parameters and dropout is inactive in evaluation mode. Equation (12) implies that a retrieved array is first averaged over its tokens and then projected, yielding one vector per modality; donor tokens are not passed to the fusion Transformer individually. The patient’s own WSI is encoded as the mean over all 128 cached rows, including zero padding, which is the historical behaviour of the frozen network; the valid-row mean of Eq. (2) is used for retrieval only. The three modality vectors are fused by the frozen Transformer encoder,

$$(\mathbf{h}_i^W, \mathbf{h}_i^R, \mathbf{h}_i^T) = \text{Transformer}_{\theta} \left((\mathbf{e}_i^W, \mathbf{e}_i^R, \mathbf{e}_i^T); \text{key-padding mask} = \neg \mathbf{f}_i \right), \quad (11)$$

with frozen parameters θ . In Eq. (13) filled modalities take part in attention as valid tokens. The output tokens are pooled with fixed weights,

$$\omega_i^m = \begin{cases} 0, & f_i^m = 0, \\ w, & v_{i,g}^m = 0, f_i^m = 1, \lambda_{c,g} = 1, \\ 1, & \text{otherwise,} \end{cases} \quad \mathbf{u}_i = \frac{\sum_m \omega_i^m \mathbf{h}_i^m}{\max(1, \sum_m \omega_i^m)}, \quad w = 0.5. \quad (12)$$

In Eq. (14) the weight $w = 0.5$ applies only to modalities filled with retrieved features ($\lambda_{c,g} = 1$). A modality filled with the training mean ($\lambda_{c,g} = 0$) has weight one, exactly as in the mean-imputation reference, and a masked modality has weight zero. Because the WSI is always valid the denominator is at least one, and the maximum mirrors the clamp of the implementation. The weight w is a hand-set constant and not a learned reliability gate, and it does not remove the influence that a filled token exerts on the other tokens through attention in Eq. (13).

Risk score and aggregation. The existing cohort-specific survival head H_c yields four discrete-time logits, from which the risk score is computed as

$$\ell_i = H_c(\mathbf{u}_i), \quad \eta_{ik} = \text{clip}(\sigma(\ell_{ik}), 10^{-6}, 1 - 10^{-6}), \quad S_{ik} = \prod_{\tau=1}^k (1 - \eta_{i\tau}), \quad R_i = - \sum_{k=1}^4 S_{ik}. \quad (13)$$

The concordance index (C-index) is computed from the risk R_i of Eq. (15), the survival time and the event indicator with the same implementation for all three strategies; no time bins, loss weights or training hyperparameters are changed. With $C_{c,g,a,s}$ the test C-index of cohort c , setting g , strategy a and seed s , the quantities reported in Table 1 are

$$\bar{C}_{c,g,a} = \frac{1}{5} \sum_{s \in \mathcal{S}} C_{c,g,a,s}, \quad O_{c,a} = \frac{1}{4} \sum_{g \in \mathcal{G}} \bar{C}_{c,g,a}, \quad \bar{C}_a = \frac{1}{5} \sum_c O_{c,a}, \quad (14)$$

where $\mathcal{S} = \{123, 132, 213, 231, 321\}$ are the seeds, $\mathcal{G}$ contains the four settings and a ranges over no compensation (m0real), mean imputation (m1) and the proposed prototype-guided compensation (strategy0). By Eq. (16) every entry for a single setting averages five seeds, the overall value $O_{c,a}$ of a cohort averages its 20 cells, and the macro-average $\bar{C}_a$ averages all 100 cells with equal weights; patients are not pooled across seeds and no seed is selected. The number of cohorts in which the proposed method exceeds both references is

$$W = \sum_c \mathbf{1}[O_{c,a^*} > O_{c,m1} \wedge O_{c,a^*} > O_{c,m0real}], \quad (15)$$

where a^* denotes the proposed method (strategy0). The comparisons in Eq. (17) use unrounded values.

3 Results

3.1 Compensation strategies under missing modalities

Table 1 compares three inference-time strategies for missing modalities on identical frozen networks: no compensation, mean imputation, and the proposed prototype-guided compensation defined in Section 2.5. Averaged with equal weights over the five cohorts, five seeds and four settings (Eq. (16)), the proposed method reached a macro-average C-index of 0.640506, compared with 0.631824 for mean imputation and 0.630751 without compensation. In the present evaluation the proposed method thus increased the average discriminative performance. Values below are given in the order proposed method, mean imputation, no compensation. The cohort-level

overall value was higher than both references in four of the five cohorts (BRCA, LGG, LUAD and UCEC; $W = 4$ in Eq. (17)), although the margin in LUAD was small (0.587039 versus 0.587004 and 0.586206), and it was not higher in BLCA (0.578982 versus 0.582191 and 0.583388). This count refers to the overall averages and does not mean that the proposed method was best in every setting. The larger differences occurred when both auxiliary modalities were masked, in BRCA (0.640263 versus 0.614738 and 0.596026), UCEC (0.609348 versus 0.575072 and 0.576319) and LGG (0.665286 versus 0.635266 and 0.576190), and for UCEC in the text-masked setting (0.609748 versus 0.577743 and 0.589272). When only RNA was masked, or without additional masking, the three strategies were close, and mean imputation or no compensation was highest in several cohorts. No statistical test was performed. [TODO: Author to add: limitation concerning how the LUAD setting was selected.]

3.2 Two-modality comparison

[TODO: Fill Table 2 from the designated sources and the existing Ours results, then describe the observed per-cohort and Overall comparisons.]

3.3 Three-modality comparison: optional

[TODO: Confirm whether Table 3 is retained and specify the third modality. If retained, fill its entries and describe the comparisons.]

4 Discussion

[TODO: Interpret the completed comparisons without inventing a mechanistic explanation or claiming an unmeasured improvement.]

[TODO: Discuss the findings in relation to the cited methods, distinguishing literature-reported values from the authors' results.]

[TODO: State the limitations supported by the final study design and completed evidence.]

5 Conclusions

[TODO: Summarize the findings and their scope after the result tables have been completed.]

Abbreviations

TCGA The Cancer Genome Atlas.

TF Transformer Fusion.

Cross-Attn Cross-Attention.

[TODO: Add the full names of the reported metric, cohort labels, and remaining method abbreviations; confirm the definition of CRAD.]

Declarations

Ethics approval and consent to participate

[TODO: Author-confirmed ethics and consent statement, including approval, exemption, or applicability details.]

Table 1: Comparison of three compensation strategies across five cancers under four modality-availability settings. Entries are test C-indices averaged over five seeds; *Overall* is the equally weighted mean of the four settings of a cohort. Proto. denotes the proposed prototype-guided compensation (Section 2.5).

Scenario	BRCA			UCEC			LUAD			BLCA			LGG		
	No comp.	Mean	Proto.	No comp.	Mean	Proto.	No comp.	Mean	Proto.	No comp.	Mean	Proto.	No comp.	Mean	Proto.
RNA masked	0.683842	0.684751	0.685705	0.647229	0.655464	0.651413	0.580660	0.580239	0.580239	0.593546	0.589105	0.588168	0.767219	0.790959	0.786611
Text masked	0.655002	0.619121	0.655002	0.589272	0.577743	0.609748	0.596630	0.595156	0.596630	0.575989	0.574913	0.575989	0.705866	0.664113	0.705866
Both masked	0.596026	0.614738	0.640263	0.576319	0.575072	0.609348	0.585469	0.588803	0.588803	0.574046	0.574774	0.561797	0.576190	0.635266	0.665286
None	0.691359	0.686068	0.688112	0.651502	0.656888	0.650968	0.582064	0.583819	0.582485	0.589972	0.589972	0.589972	0.796825	0.799517	0.797723
Overall	0.656557	0.651170	0.667270	0.616081	0.616292	0.630369	0.586206	0.587004	0.587039	0.583388	0.582191	0.578982	0.711525	0.722464	0.738872

Scenarios: RNA masked, Text masked and Both masked remove the RNA profile, the pathology report or both for all test patients (rna_100, text_100, both_100); None (none) adds no artificial mask but retains naturally occurring missingness and is therefore not a complete-data condition. Overall is an average and not a fifth experimental scenario. Strategies: No comp., no compensation (m0real), missing modalities are masked; Mean, mean imputation (m1), a missing modality is replaced by the element-wise training mean of the same cohort; Proto., the proposed prototype-guided compensation (strategy0; Section 2.5). Bold marks the highest unrounded value among the three strategies within a cohort and row; ties are all bold. The three strategies use the same 25 frozen networks; no model was retrained and no statistical test was performed.

Table 2: Two-modality comparison using text and genomics. Metric and uncertainty summary: TODO.

Method	BLCA	BRCA	GBMLGG	LUAD	UCEC	CRAD	HNSC	STAD	Overall
Cross-Attn Fusion	–	–	–	–	–	–	–	–	–
Early Fusion	–	–	–	–	–	–	–	–	–
Late Fusion	–	–	–	–	–	–	–	–	–
Ours	–	–	–	–	–	–	–	–	–

Non-Ours values are planned to be taken from Table 1 of Song et al. [?], not reproduced in this study. Ours values will be supplied from the authors’ own experiments. The symbol – denotes an unfilled entry. Overall: TODO, including whether it is transcribed or calculated.

Table 3: Optional three-modality comparison using text, genomics, and a third modality (TODO). Metric and uncertainty summary: TODO.

Method	BLCA	BRCA	GBMLGG	LUAD	UCEC	CRAD	HNSC	STAD	Overall
LIMOE	–	–	–	–	–	–	–	–	–
MAGGate	–	–	–	–	–	–	–	–	–
MulT	–	–	–	–	–	–	–	–	–
TF	–	–	–	–	–	–	–	–	–
Cross-Attn Fusion	–	–	–	–	–	–	–	–	–
Early Fusion	–	–	–	–	–	–	–	–	–
Late Fusion	–	–	–	–	–	–	–	–	–
Ours	–	–	–	–	–	–	–	–	–

Non-Ours values are planned to be taken from Table 1 of Song et al. [?], not reproduced in this study. Ours values will be supplied from the authors’ own experiments. TF denotes Transformer Fusion. The symbol – denotes an unfilled entry. Overall: TODO, including whether it is transcribed or calculated.

Consent for publication

[TODO: Author-confirmed consent-for-publication statement.]

Availability of data and materials

[TODO: Dataset access details and links to available code, split files, and result records. State any actual access restrictions.]

Competing interests

[TODO: Author-confirmed declaration of competing interests.]

Funding

[TODO: Funding bodies, grant identifiers, and funder roles, or an author-confirmed statement of no funding.]

Authors' contributions

[TODO: Author names or initials and their actual contributions.]

Acknowledgements

[TODO: Acknowledgements and any required disclosure of writing assistance, after author confirmation.]